

Introduction to R | *Introduction à R*

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Section 1

Guide

- Objects
- Functions

- Objets
- Fonctions

Section 2

Getting started

This section is for readers opening R for the first time. It covers the practical setup the rest of the course assumes.

Cette section s'adresse à ceux qui ouvrent R pour la première fois. Elle couvre la configuration pratique que le reste du cours présuppose.

R is the programming language and computational engine.

RStudio is a separate application that provides a more comfortable interface to R.

Install R first, then RStudio.

R est le langage et le moteur de calcul.

RStudio est une application distincte qui offre une interface plus confortable à R.

Installez R d'abord, puis RStudio.

RStudio panes

Les fenêtres de RStudio

When you open RStudio you typically see four panes:

- **Console** — run commands one at a time; results print here
- **Source / Editor** — write scripts (.R) or Quarto (.qmd); code runs only when you send it

À l'ouverture de RStudio, vous voyez typiquement quatre fenêtres :

- **Console** — exécuter des commandes une par une ; les résultats s'affichent ici
- **Source / Éditeur** — écrire des scripts (.R) ou Quarto (.qmd) ; le code ne s'exécute que lorsque vous l'envoyez

RStudio panes (2)

Les fenêtres de RStudio (2)

- **Environment / History** — objects in memory and command history
 - **Files / Plots / Help / Packages** — file browser, plots, help pages, installed packages
- **Environnement / Historique** — objets en mémoire et historique des commandes
 - **Fichiers / Graphiques / Aide / Packages** — explorateur de fichiers, graphiques, aide, packages installés

Running code

Exécuter du code

Type in the **console** and press Enter, or write in a **script** and send with **Ctrl + Enter** (Windows/Linux) or **Cmd + Enter** (Mac). Scripts are preferable — work is saved and reproducible.

Tapez dans la **console** et appuyez sur Entrée, ou écrivez dans un **script** et envoyez avec **Ctrl + Entrée** (Windows/Linux) ou **Cmd + Entrée** (Mac). Les scripts sont préférables — le travail est sauvegardé et reproductible.

Assignment vs display

Affectation vs affichage

Two kinds of commands behave differently:

Deux types de commandes se comportent différemment :

```
2 + 2          # add two numbers and show the result
```

```
[1] 4
```

```
x <- 2 + 2      # compute 2 + 2 and store in object x
```

```
x              # show the value stored in x
```

```
[1] 4
```

Working directory

Répertoire de travail

The **working directory** is the folder R reads from and writes to by default. Many early errors come from R looking in the wrong place for a data file.

Le **répertoire de travail** est le dossier que R utilise par défaut pour lire et écrire. Beaucoup d'erreurs viennent d'un mauvais chemin vers un fichier de données.

```
getwd()                # current working directory
```

```
[1] "C:/WZB Dropbox/Macartan Humphreys/LDID 2026/Slides_abidjan_2026"
```

```
list.files("data/")    # files R can see in a folder
```

```
[1] "echocardiogram.data" "ships.csv"           "ships_out.csv"
```

Best practice: work inside an **RStudio Project**.

- ➊ File → New Project → New Directory → New Project
- ➋ Put scripts and data inside the project folder
- ➌ Open the project each session — `getwd()` should point to the project root

Bonne pratique : travailler dans un **projet RStudio**.

- ➊ Fichier → Nouveau projet → Nouveau répertoire → Nouveau projet
- ➋ Placer scripts et données dans le dossier du projet
- ➌ Ouvrir le projet à chaque session — `getwd()` doit pointer vers la racine du projet

R's base installation is extended by **packages**.
Install once; **load** each session with `library()`.

L'installation de base de R est étendue par des **packages**. **Installer** une fois ; **charger** à chaque session avec `library()`.

```
install.packages("readr")    # download and install (once)
library(readr)               # load for this session
```

Getting help

Obtenir de l'aide

Read error messages from the **bottom up** — they usually identify the object or file that caused the problem.

Lisez les messages d'erreur **de bas en haut** — ils identifient en général l'objet ou le fichier en cause.

```
?mean          # help page for mean()
args(mean)     # arguments that mean() accepts
```

Housekeeping

Nettoyage de la session

```
ls()                                # list objects in memory
```

```
[1] "x"
```

```
rm(x)                              # remove object x
```

```
rm(list = ls())                    # remove all objects (use with care!)
```

Section 3

Scalars, vectors, and arithmetic

R as a calculator

R comme calculatrice

R works as a calculator. Assignment uses `<-` (or `=`). Common math functions are built in.

R fonctionne comme une calculatrice. L'affectation utilise `<-` (ou `=`). Les fonctions mathématiques courantes sont intégrées.

```
r <- 3  
pi * r^2                # area of a circle
```

```
[1] 28.27433
```

```
sin(pi / 4)
```

```
[1] 0.7071068
```

```
sqrt(2)
```

```
[1] 1.414214
```

```
log(2)
```

```
[1] 0.6931472
```

Creating vectors

Créer des vecteurs

Vectors are created with `c()`. Arithmetic is **element-wise**.

Les vecteurs se créent avec `c()`. L'arithmétique est **élément par élément**.

```
x <- c(2, 4, 6, -1)
```

```
y <- c(4, 1, 2, 2)
```

```
x + y
```

```
[1] 6 5 8 1
```

```
x - y
```

```
[1] -2 3 4 -3
```

```
x / y
```

```
[1] 0.5 4.0 3.0 -0.5
```

```
x * y
```

```
[1] 8 4 12 -2
```

Vector summaries

Résumés d'un vecteur

```
length(x)
```

```
[1] 4
```

```
mean(x)
```

```
[1] 2.75
```

```
sd(x)
```

```
[1] 2.986079
```

```
var(x)
```

```
[1] 8.916667
```

```
z <- c("A", "B", "a", "c")
```

```
z[1]
```

```
[1] "A"
```

```
z[2:3]
```

Section 4

Data frames

What is a data frame?

Qu'est-ce qu'un data frame ?

Most real data lives in **data frames**: tables of rows (cases) and columns (variables). Columns can differ in type.

La plupart des données sont des **data frames** : tableaux de lignes (cas) et colonnes (variables). Les colonnes peuvent être de types différents.

```
height <- c(172, 165, 180, 158)
weight <- c(70, 60, 82, 55)
sex     <- c("F", "F", "M", "F")

people <- data.frame(height, weight, sex)
people
```

	height	weight	sex
1	172	70	F
2	165	60	F
3	180	82	M
4	158	55	F

Accessing columns

Accéder aux colonnes

```
people$height
```

```
[1] 172 165 180 158
```

```
people[, "height"]
```

```
[1] 172 165 180 158
```

```
people[, 1]
```

```
[1] 172 165 180 158
```

```
people[1:2, ]
```

	height	weight	sex
1	172	70	F
2	165	60	F

```
people[1:2, 1]
```

```
[1] 172 165
```

Section 5

Reading data

Reading a CSV

Lire un CSV

Put the data file in your project folder, then use `read.csv()` or `readr::read_csv()`.

Placez le fichier dans le dossier du projet, puis utilisez `read.csv()` ou `readr::read_csv()`.

```
ships <- read.csv("data/ships.csv")
```

```
class(ships)
```

```
[1] "data.frame"
```

```
names(ships)
```

```
[1] "Type"          "Construction" "Operation"     "Months"       "Dammage"
```

```
head(ships)
```

	Type	Construction	Operation	Months	Dammage
1	A	60	60	127	0
2	A	60	75	63	0
3	A	65	60	1095	3
4	A	65	75	1095	4
5	A	70	60	1512	6

Missing values on import

Valeurs manquantes à l'import

```
Echo <- read.csv("data/echocardiogram.data",  
                header      = FALSE,  
                na.strings = c("?"))  
  
names(Echo) <- c("survival", "alive", "age", "peri",  
                "frac_short", "epss", "lvdd")  
  
head(Echo)
```

	survival	alive	age	peri	frac_short	epss	lvdd	<NA>	<NA>	<NA>	<NA>	<NA>
1	11	0	71	0	0.260	9.000	4.600	14	1.00	1.000	name	1
2	19	0	72	0	0.380	6.000	4.100	14	1.70	0.588	name	1
3	16	0	55	0	0.260	4.000	3.420	14	1.00	1.000	name	1
4	57	0	60	0	0.253	12.062	4.603	16	1.45	0.788	name	1
5	19	1	57	0	0.160	22.000	5.750	18	2.25	0.571	name	1
6	26	0	68	0	0.260	5.000	4.310	12	1.00	0.857	name	1
<NA>												
1	0											

Writing data

Exporter des données

```
write.csv(ships,  
          file = "ships_out.csv",  
          row.names = FALSE)
```

Section 6

Data types

Factors

Facteurs (variables catégorielles)

Categorical variables are stored as **factors**.
Inspect levels with `levels()` and counts with `table()`.

Les variables catégorielles sont des **facteurs**.
Inspectez les niveaux avec `levels()` et les effectifs avec `table()`.

```
ships$Type      <- as.factor(ships$Type)
ships$Operation <- as.factor(ships$Operation)
```

```
levels(ships$Operation)
```

```
[1] "60" "75"
```

```
table(ships$Operation)
```

```
60 75
20 20
```

Re-leveling factors

Recoder les niveaux

Use `relevel()` to change the **reference category**.

Utilisez `relevel()` pour changer la **catégorie de référence**.

```
ships$Operation <- relevel(ships$Operation, ref = "75")  
levels(ships$Operation)
```

```
[1] "75" "60"
```

Factor codes vs labels

Codes vs étiquettes

Warning: `as.numeric()` on a factor returns underlying **codes**, not the labels.

Attention : `as.numeric()` sur un facteur renvoie les **codes** internes, pas les étiquettes.

```
codes <- as.numeric(ships$Operation)
labels <- as.character(ships$Operation)
```

```
codes[1:5]
```

```
[1] 2 1 2 1 2
```

```
labels[1:5]
```

```
[1] "60" "75" "60" "75" "60"
```

Section 7

Summary statistics

summary()

summary()

summary() gives an overview of each variable in a data frame.

summary() donne un aperçu de chaque variable d'un data frame.

```
summary(ships)
```

Type	Construction	Operation	Months	Dammage
A:8	Min. :60.00	75:20	Min. : 0.0	Min. : 0.0
B:8	1st Qu.:63.75	60:20	1st Qu.: 175.8	1st Qu.: 0.0
C:8	Median :67.50		Median : 782.0	Median : 2.0
D:8	Mean :67.50		Mean : 4089.3	Mean : 8.9
E:8	3rd Qu.:71.25		3rd Qu.: 2078.5	3rd Qu.:11.0
	Max. :75.00		Max. :44882.0	Max. :58.0

Numeric summaries

Résumés numériques

```
mean.dam <- mean(ships$Dammage)
sd.dam   <- sd(ships$Dammage)
var.dam  <- var(ships$Dammage)

quantile(ships$Dammage,
         probs = c(0.25, 0.5, 0.75))
```

25%	50%	75%
0	2	11

Covariance and correlation

Covariance et corrélation

```
cov(ships$Months, ships$Damage)
```

```
[1] 115306.1
```

```
cor(ships$Months, ships$Damage)
```

```
[1] 0.8525174
```

Tables and missing values

Tableaux et valeurs manquantes

```
table(ships$Type)
```

```
A B C D E  
8 8 8 8 8
```

```
table(ships$Operation, ships$Type)
```

```
      A B C D E  
75 4 4 4 4 4  
60 4 4 4 4 4
```

```
mean(Echo$age)
```

```
[1] NA
```

```
mean(Echo$age, na.rm = TRUE)
```

```
[1] 62.81372
```

Section 8

Plotting with ggplot2

The ggplot pattern

Le modèle ggplot

We use **ggplot2** (tidyverse). Basic pattern:

Nous utilisons **ggplot2** (tidyverse). Modèle de base :

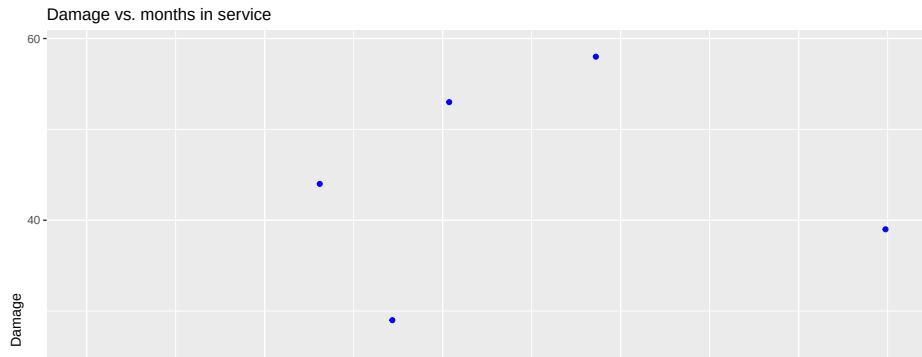
```
ggplot(data = ..., aes(x = ..., y = ...)) +  
  geom_...()
```

```
library(ggplot2)
```

Scatterplot

Nuage de points

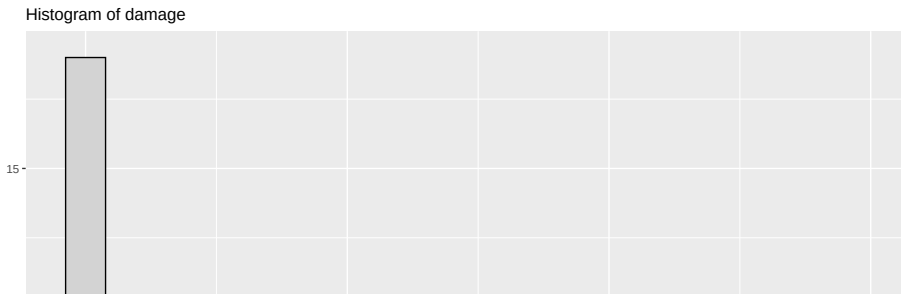
```
ggplot(data = ships,  
       aes(x = Months, y = Dammage)) +  
  geom_point(color = "blue") +  
  labs(x = "Months",  
       y = "Damage",  
       title = "Damage vs. months in service")
```



Histogram

Histogramme

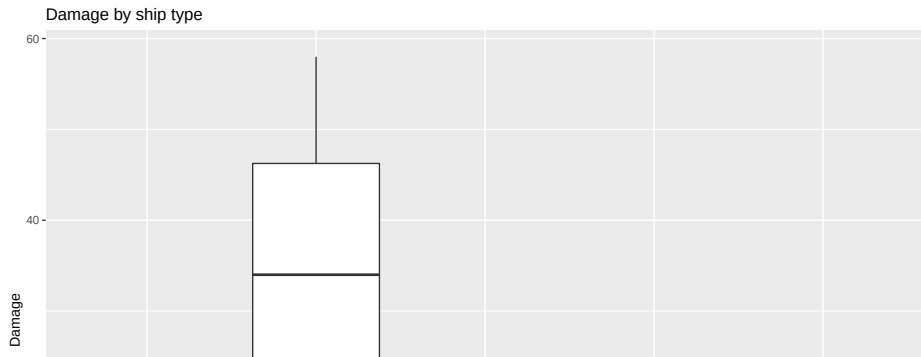
```
ggplot(data = ships,  
       aes(x = Dammage)) +  
  geom_histogram(bins = 20,  
                fill = "lightgray",  
                color = "black") +  
  labs(x = "Damage",  
       y = "Count",  
       title = "Histogram of damage")
```



Boxplot by group

Boîte à moustaches par groupe

```
ggplot(data = ships,  
       aes(x = Type, y = Dammage)) +  
  geom_boxplot() +  
  labs(x = "Ship type",  
       y = "Damage",  
       title = "Damage by ship type")
```



- 1 Open the RStudio project
- 2 Create a script `intro.R`
- 3 Read the data
- 4 Inspect variables
- 5 Summarize and plot with `ggplot`

- 1 Ouvrir le projet RStudio
- 2 Créer un script `intro.R`
- 3 Importer les données
- 4 Inspecter les variables
- 5 Résumer et graphiquer avec `ggplot`

Mini workflow (code)

Mini parcours (code)

```
ships <- read.csv("data/ships.csv")
```

```
str(ships)
```

```
'data.frame':  40 obs. of  5 variables:
 $ Type      : chr  "A" "A" "A" "A" ...
 $ Construction: int   60 60 65 65 70 70 75 75 60 60 ...
 $ Operation  : int   60 75 60 75 60 75 60 75 60 75 ...
 $ Months     : int  127 63 1095 1095 1512 3353 0 2244 44882 17176 ...
 $ Damage     : int   0 0 3 4 6 18 0 11 39 29 ...
```

```
summary(ships$Damage)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.0	0.0	2.0	8.9	11.0	58.0

```
ggplot(data = ships,
       aes(x = Months, y = Damage)) +
  geom_point(color = "darkred") +
  labs(x = "Months in service")
```

Section 9

Let's go

Simulation by hand

Simulation à la main

A tiny randomized experiment in base R:

Une petite expérience randomisée en R de base :

```
N <- 10
Z <- sample(0:1, N, replace = TRUE)
Y <- Z + rnorm(N)

lm(Y ~ Z) |> summary()
```

Call:

```
lm(formula = Y ~ Z)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.2686	-0.8483	-0.3718	0.7068	1.8617

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.5205	0.6216	0.837	0.427

Same in DeclareDesign

Même chose avec DeclareDesign

The same design declared with **DeclareDesign**:

La même conception déclarée avec
DeclareDesign :

```
library(DeclareDesign)
```

```
N <- 10
design <-
  declare_model(N,
    Z = sample(0:1, N, replace = TRUE),
    Y = Z + rnorm(N)) +
  declare_inquiry(ATE = 1) +
  declare_estimator(Y ~ Z)

design
```

Research design declaration summary

Step 1 (model): declare_model(N, Z = sample(0:1, N, replace = TRUE), Y = Z + rnorm